

Complete Git, GitHub & Windows BAT Notes

Detailed explanations of Git/GitHub commands, workflows, branching, remotes, undoing changes, and common Windows batch commands.

git init

Command:

```
git init
```

Explanation: Creates a new Git repository. Git creates a hidden .git directory containing repository data such as history, references, configuration, and objects.

git clone

Command:

```
git clone https://github.com/user/project.git
```

Explanation: Downloads an existing repository, its files, and its Git history. It also normally creates a remote named origin.

git status

Command:

```
git status
```

Explanation: Shows the current branch, staged changes, unstaged changes, and untracked files.

git add

Command:

```
git add app.py  
git add .
```

Explanation: Moves selected changes into the staging area. The staging area determines what the next commit will contain.

git commit

Command:

```
git commit -m "Added login page"
```

Explanation: Creates a saved snapshot of staged changes. A commit has an ID, author, timestamp, message, and parent history.

git log

Command:

```
git log  
git log --oneline
```

Explanation: Displays commit history. --oneline gives a compact one-line view.

git diff

Command:

```
git diff
```

Explanation: Shows unstaged differences so you can review changes before staging them.

git branch

Command:

```
git branch  
git branch feature
```

Explanation: Lists branches or creates a new branch.

git checkout

Command:

```
git checkout feature  
git checkout -b feature
```

Explanation: Switches branches. -b creates and switches to a new branch. For ordinary branch switching, git switch is now clearer.

git switch

Command:

```
git switch feature  
git switch -c feature
```

Explanation: Modern branch-switching command. -c creates and switches to a branch.

git merge

Command:

```
git switch main  
git merge feature
```

Explanation: Combines feature into the currently checked-out branch. It may fast-forward or create a merge commit.

git remote

Command:

```
git remote -v
```

Explanation: Shows configured remote names and URLs.

git remote add

Command:

```
git remote add origin https://github.com/user/project.git
```

Explanation: Connects the local repository to GitHub. origin is the conventional remote name.

git remote remove

Command:

```
git remote remove origin
```

Explanation: Removes the local remote configuration; it does not delete the GitHub repository.

git remote set-url

Command:

```
git remote set-url origin NEW_URL
```

Explanation: Changes the URL of an existing remote, useful when a repository moves or when changing HTTPS/SSH.

git push

Command:

```
git push origin main
```

Explanation: Uploads local commits from main to the origin remote.

git pull

Command:

```
git pull origin main
```

Explanation: Gets remote changes and integrates them into the current branch. It is commonly fetch plus merge, or fetch plus rebase depending on configuration/options.

git fetch

Command:

```
git fetch
```

Explanation: Downloads remote updates without integrating them into your current branch, allowing you to inspect them first.

git stash

Command:

```
git stash  
git stash pop
```

Explanation: Temporarily saves uncommitted changes. stash pop restores the latest stash and removes it if restoration succeeds.

git reset

Command:

```
git reset --soft HEAD~1  
git reset --hard HEAD~1
```

Explanation: Moves HEAD backward. --soft keeps changes staged. --hard also resets the index and working tree and can permanently discard uncommitted changes.

git restore

Command:

```
git restore app.py
```

Explanation: Restores a file from the appropriate Git state, commonly discarding its unstaged changes.

git rm

Command:

```
git rm app.py
```

Explanation: Deletes a tracked file and stages that deletion for the next commit.

git mv

Command:

```
git mv old.py new.py
```

Explanation: Renames or moves a tracked file while updating Git's index.

git tag

Command:

```
git tag v1.0  
git push origin v1.0
```

Explanation: Creates a named reference to a commit, commonly used to mark releases.

git config

Command:

```
git config --global user.name "Swetanshu"  
git config --global user.email "abc@gmail.com"  
git config --list
```

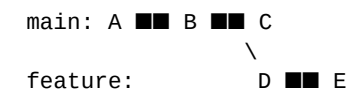
Explanation: Sets or displays Git configuration. --global applies the setting to your user account.

Complete GitHub First-Push Workflow

```
git init
git add .
git commit -m "Initial Commit"
git branch -M main
git remote add origin https://github.com/user/project.git
git push -u origin main
```

- 1. git init — initializes Git.
- 2. git add . — stages project files.
- 3. git commit — creates the first local snapshot.
- 4. git branch -M main — renames the current branch to main.
- 5. git remote add origin — connects the local repository to GitHub.
- 6. git push -u origin main — uploads main and sets origin/main as its upstream.

Branching and Merge Example



Running git merge feature while on main combines the feature history into main. If histories diverged, Git may create a merge commit.

Common Windows BAT Commands

Command	Purpose
@echo off	Hides commands as they execute.
echo Hello	Prints text.
pause	Waits for a key press.
cls	Clears the console.
title My Script	Sets the console title.
color 0A	Changes console colors.
cd folder	Changes directory.
dir	Lists files and folders.
mkdir Test	Creates a directory.
rmdir Test	Removes a directory.
del file.txt	Deletes a file.
copy a.txt b.txt	Copies a file.
move a.txt Backup\	Moves a file.
ren old.txt new.txt	Renames a file.
set NAME=Swetanshu	Sets an environment variable.
if exist file.txt	Conditional check for a file.
goto LABEL	Jumps to a labeled section.
call script.bat	Runs another batch file and returns.
start notepad	Starts a program.
tasklist	Lists running processes.

Command	Purpose
taskkill /IM notepad.exe /F	Forces a process to stop.
ipconfig	Displays network configuration.
ping google.com	Tests connectivity.
shutdown /s /t 0	Shuts down Windows immediately.
exit	Closes the command shell.

Git Automation BAT Example

```
@echo off
echo Adding files...
git add .

echo Creating commit...
git commit -m "Daily update"

echo Uploading to GitHub...
git push origin main

echo Done!
pause
```

This script stages changes, creates a commit, pushes main to origin, prints progress, and pauses at the end.

Git Concepts Cheat Sheet

Concept	Meaning
Working Directory	Your files as you edit them.
Staging Area	Changes selected for the next commit.
Commit	Saved snapshot of staged changes.
Branch	Movable reference used for independent development.
HEAD	Reference to the currently checked-out commit/branch.
Remote	Named connection to another repository.
origin	Conventional name for the default remote.
main	Common name for the primary branch.
Upstream	Remote branch associated with a local branch.

Quick Git Cheat Sheet

```
git init # initialize repository
git clone URL # clone repository
git status # check status
git add . # stage all changes
git add file.py # stage one file
git commit -m "message" # create commit
git log --oneline # compact history
git branch # list branches
git switch -c feature # create + switch branch
git switch main # switch branch
git merge feature # merge feature into current branch
git remote -v # show remotes
git remote add origin URL # add GitHub remote
git fetch # download remote updates
git pull origin main # fetch + integrate
git push origin main # upload commits
git stash # temporarily store changes
```

```
git stash pop # restore stash
git restore file.py # restore file
git reset --soft HEAD~1 # undo commit, keep changes staged
git reset --hard HEAD~1 # undo commit and working changes
git tag v1.0 # create tag
git push origin v1.0 # push tag
```